An Interpretable Hybrid AI Framework for Medical Diagnosis with Transformer-Based Comparative Analysis

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***Abstract*—The problem of diagnosing a disease in the field of medicine is quite a difficult task. Even though modern-day machine learning models perform better when it comes to prediction accuracy, they lack transparency and function as black boxes, which makes them unsuitable for application in certain industries, such as medicine. This paper presents a novel approach of using an explainable hybrid system for medical diagnoses by integrating knowledge-based intelligent agents and probability theory. A hybrid decision-making approach with conflict resolution and Bayesian override mechanism is proposed in order to improve the robustness of the decision-making process. In addition to that, an explanation component creates reasoning trails in order to provide more clarity. Plus, we implemented a Transformer-based deep learning model in order to compare probabilistic and symbolic approaches and using accuracy, F1-score, precision, recall, ROC-AUC analysis, confusion matrices, and training loss visualization. There is a balanced tradeoff between interpretability and diagnostic accuracy.**

***Index Terms*—Explainable AI, Medical Diagnosis, Transformer Networks, Bayesian Inference, Hybrid AI, Deep Learning, Rule-Based Systems**

I. Introduction

Medical diagnosis problem is based on the ability to detect the set of possible diseases by linking up the symptoms of a patient in the environment of uncertainty. Traditional clinical decision-making strategies use large amounts of expertise and comprehensible algorithms. However, modern advances in artificial intelligence technologies allow machine learning algorithms to achieve significantly better performance in prediction without using interpretable methods. Such approaches cause some challenges in the medical industry where accountability and justification are considered critical aspects of decision-making. Therefore, to deal with such problems, we introduce the idea of developing a hybrid explainable approach, which will combine logic reasoning and probabilistic analysis techniques.

II. Related Work

Approaches like MYCIN showed that rule-based expert systems could effectively perform medical diagnosis. However, they failed to handle uncertainty and scalability problems. The machine learning algorithms like Naive Bayes and Deep Neural Networks proved to work better for predictions, but they lacked transparency. The current state of XAI research solves this problem. Moreover, there are approaches that combine symbolic and probabilistic techniques; however, only a few systems consider conflict resolution explicitly in their explainability process. More recently, Transformer-based neural architectures have become popular because of their strong performance in healthcare analytics, disease prediction, and clinical decision support systems. However, these methods often lack clear reasoning pathways, which makes them hard to use in high-accountability medical settings. This research tries to address this issue by combining rule-based inference, Bayesian probabilistic reasoning, and Transformer-based deep learning in a single explainable framework for comparative medical diagnosis analysis.

The current research is attempting to build a unified explainable framework for comparative medical diagnosis analysis and solve the problem by integrating rule-based inference, Bayesian probabilistic reasoning, and Transformer-based deep learning models.

III. System Overview

The proposed system is based on three main modules:

- Rule-Based Inference Engine
- Bayesian Probabilistic Model
- Hybrid Decision Controller

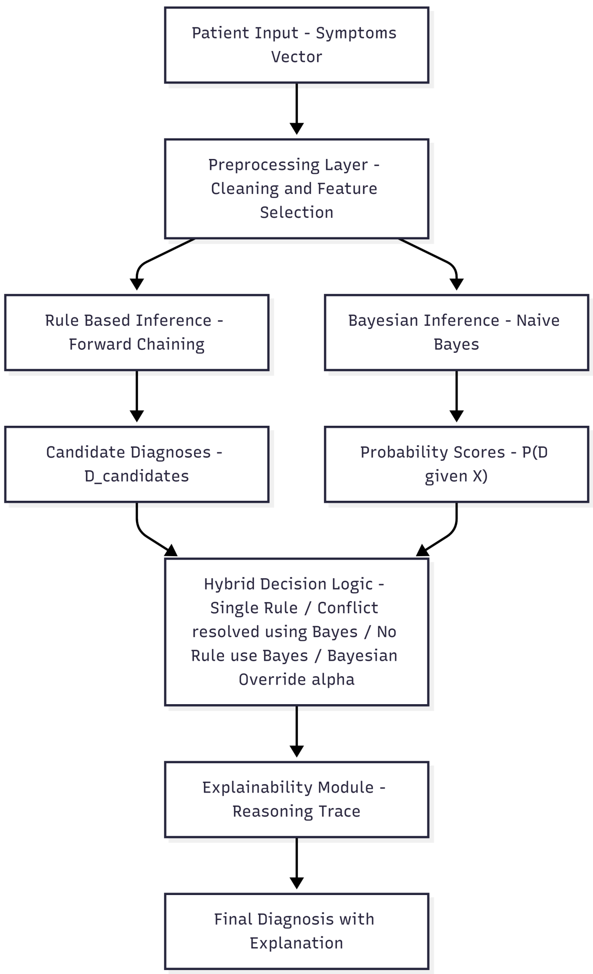

Fig. 1. Proposed Hybrid System Architecture

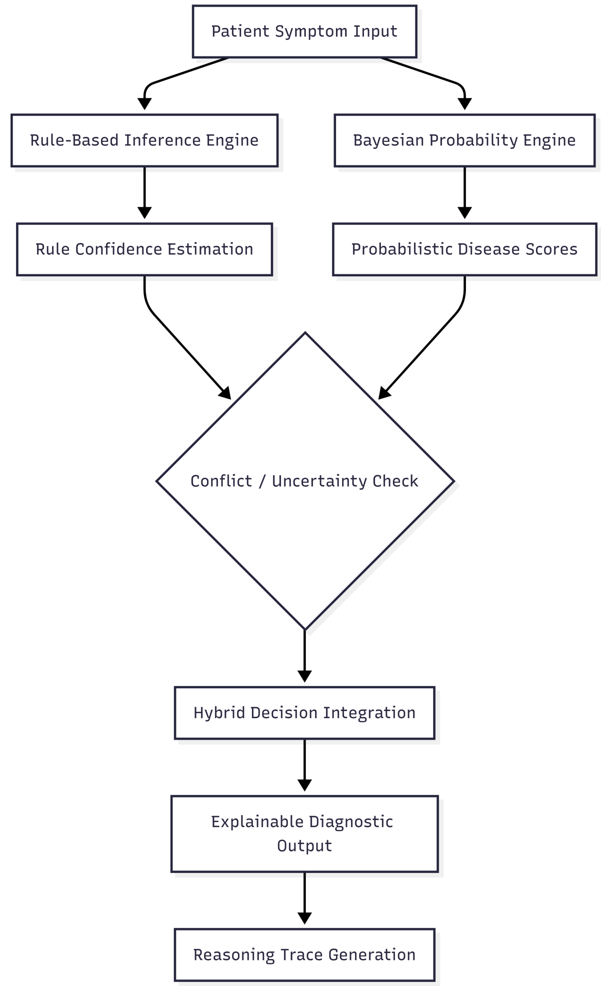

Fig. 2. Explainable Hybrid Diagnostic Framework

IV. Dataset and Preprocessing

The data used in this research is a set of medical records that use binary symptom vector representations of patients. Each attribute describes specific symptoms, with the value of 1 corresponding to presence and 0 to absence of these symptoms.

Disease diagnosis acts as a target variable. Initially, there were 246,945 examples with 378 symptoms. When using a lot of dimensions within a dataset, it causes inefficiency during processing and negatively impacts the performance of predictive models. Thus, preprocessing of the dataset becomes a requirement.

A. Data Cleaning and Deduplication

At first step, we eliminated all possible duplications from the data. It was needed to avoid introducing any biases during training due to duplicated information. The number of cases within the data set was initially 246,945 and was trimmed down to 189,647.

B. Feature Pruning

We found that various attributes in the datasets had zero variance that means they took same value for all the observations which in turn just increased the computational load because they were of no use in our result evaluation. Hence, they were dropped, which resulted in a reduction of the number of variables from 378 to 329.

C. Class Filtering and Balancing

Initially , the dataset contained many diseases with uneven class distribution. In order, to maintain diversity within classes and make a balanced classification problem, we took only the top 15 most frequent diseases. As a result, our dataset shrank, with only 18,171 entries remaining.

D. Feature Selection

To further reduce dimensionality, we used a frequency-based feature selection technique to keep only the important features.

We retained features that occurred frequently for the selected diseases, while less important features were removed, resulting in a final selection of 40 symptoms. This reduction not only improves computation efficiency but also enhances the clarity of interpretation.

E. Data Representation

The post-processing step creates a patient profile characterized by a 40-bit binary vector and a label for the patient's disease. This concise representation is suitable for both rule-based and probabilistic inference. To summarize, we would write that the preprocessing process makes the data clean, balanced and easy to compute, with keeping the relevant information for rule based and Bayesian analysis.

V. Rule-Based Inference

Now we will make rule based inferences in order to encode to knowledge about domain into logical rules and performing deterministic reasoning based on patients' symptoms. This part forms the symbolic part of the hybrid system, that ensures interpretability by clearly mapping symptoms to diseases.

A. Knowledge Representation

We represented medical knowledge using production rules in the following form:

$$IF\ (x_1 \wedge x_2 \wedge \cdots \wedge x_n) \rightarrow D_k$$

where xi indicates the presence of the ith symptom and Dk signifies the associated disease. Each rule defines a medically relevant relationship that can be extracted from the data. Initially, the symptoms present in the patient were mapped to logical facts which, helps us effectively to match the patient data with the rules. Mathematically, a patient's case can be expressed by the formula:

$$F = \{x_i \mid x_i = 1\}$$

where the set of observed symptoms is denoted by F.

B. Forward Chaining Mechanism

The reasoning technique adopted in the inferencing process is known as forward chaining. In this type of inference, the starting point includes information given as well as symptoms exhibited by the patient, with an objective of inferring possible results using the rules.

In the reasoning process, each of the rules will be tried on the symptoms provided based on whether all the symptoms required by the rule are satisfied in the given symptoms of the patient.

```
for conditions, disease in rules:
    if all(sym in facts for sym in conditions):
        diagnoses.append(disease)
```

Such a system can incorporate all relevant regulations into consideration and, thus, come up with a wide range of possible conclusions.

C. Deterministic Reasoning and Interpretability

Among the strengths of the rule-based inferencing process, one can highlight its deterministic character. For any predetermined symptoms and rule base, the output will be consistent and the same for all cases. Such determinism is important when we use inferencing for medical purposes.

Another important feature of this approach is its transparency. Every diagnosis will be derived from the specific rule that causes it. Thus, it is possible to develop an explanation like:

- Rule R1 fired: fever and cough imply pneumonia
- Rule R2 fired: cough and sore throat imply strep throat

In these situations, tracking information helps health professionals understand and confirm the decision-making process.

D. Limitations of Rule-Based Inference

However, despite the benefits mentioned above, this method has some basic limitations. The rule-based approach operates only on rules that were set from the beginning. It cannot handle uncertainties or partial matches of symptoms. When multiple rules are activated at once, the outcome becomes unclear because several possible diagnoses are generated.

Additionally, when no rule fits the symptoms provided by a user, the diagnostic process cannot work. So, it is necessary to introduce a probabilistic model into the system.

VI. Conflict Detection

It should be noted that when it comes to diagnosis in the real world, there may be many diseases that share similar symptoms. Therefore, it becomes evident that during matching between the symptom vector of the patient and the rule base, there will be several rules that are simultaneously found to be true.

Let F represent the set of symptoms exhibited by a patient, while the rule base is assumed to consist of a set of production rules relating symptom vectors to diseases. The output of a forward-chaining operation will be the following set of diagnoses:

$$D_{\text{candidates}} = \{D_1, D_2, \ldots, D_m\}$$

where each D_i represents a disease for which all the rule conditions can be met from the symptom set F.

A. Definition of Conflict

A conflict exists when there is more than one possible diagnosis, that is,

$$|D_{\text{candidates}}| > 1$$

This means that various diseases can be equally possible based on the symptoms given in relation to the rule-based approach. This is because the rule-based approach is deterministic in nature and therefore cannot rank any particular disease higher than the other.

B. Cause of Conflict

The main cause of conflict is the vagueness of medical data. There are some symptoms found in most diseases like fever, fatigue, and coughs. Therefore, in certain situations, two or more rules may be activated for a single patient because the same criteria apply to both rules.

This situation can be seen in the following example. A patient having fever and coughing may be diagnosed with different types of illnesses like pneumonia, influenza, and bronchitis. Thus, more than one rule may produce an accurate output.

C. Detection Mechanism

It directly identifies any conflict based on the size of the possible diagnoses in the process of reasoning. This works using the following algorithm:

```
if len(diagnoses) > 1:
    trace.append("Conflict detected")
```

This stage is important for enabling hybrid inference, as it signals when probabilistic reasoning needs to be applied.

D. Implications of Conflict

A conflict represents one of the weaknesses of symbolic reasoning. In reality, even though rule-based reasoning yields understandable results, it fails to handle uncertainty associated with medical diagnostics.

Instead of solving a conflict through random choice, the suggested solution involves the use of Bayes' probabilistic reasoning for sorting out the most probable disease. Thus, the decision becomes easy to understand and scientifically sound.

To conclude, it should be noted that conflict detection allows for bridging the gap between two types of reasoning employed in the suggested hybrid approach.

VII. Bayesian Inference

Moreover, the Bayesian inference procedure will be included in the proposed framework in order to deal with the uncertainty related to the diagnosis of diseases from the symptoms that have been noted. In this regard, while the rule-based procedure yields conclusive results, the Bayesian inference procedure yields probability values for all possible diseases.

A. Problem Formulation

Let $X = x_1, x_2, \ldots, x_n$ be the observed symptoms of a patient, where each x_i is a binary variable. This variable takes a value of 1 if the symptom is present and 0 if absent. Let us take D as the disease of interest, which is, chosen from several diseases.

We need to calculate the posterior probability as:

$$P(D \mid X)$$

This represents the probability of disease D given the observed symptoms X.

B. Naive Bayes Assumption

According to Bayes' theorem, posterior probability is calculated as follows:

$$P(D \mid X) = \frac{P(X \mid D) \cdot P(D)}{P(X)}$$

Given that $P(X)$ is fixed across diseases, it does not affect the comparison. Hence, the new criterion is:

$$P(D \mid X) \propto P(D) \cdot P(X \mid D)$$

In order to make calculations easier, an assumption known as Naive Bayes is considered. According to it, all symptoms are independent given the disease. As a result, the expression for the likelihood factorization takes the form:

$$P(X \mid D) = \prod_{i=1}^{n} P(x_i \mid D)$$

Consequently, the posterior probability will equal:

$$P(D \mid X) = P(D) \prod_{i=1}^{n} P(x_i \mid D)$$

C. Parameter Estimation

In the dataset, prior probability $P(D)$ is determined as the relative frequency of disease D :

$$P(D) = \frac{\text{count}(D)}{N}$$

Here, N denotes the size of the sample. Conditional probability is computed as follows:

$P(x_i \mid D)$

This term describes the probability of symptoms x_i if disease D exists. In this case, the probability value will be calculated based on frequency data of the dataset. For binary attributes:

$$P(x_i = 1 \mid D) = \frac{\text{count}(x_i = 1, D)}{\text{count}(D)}$$

In case where we need to avoid zero probability, Laplace smoothing technique was applied. Basically, the smoothing mechanism improved numerical stability to prevent unseen symptom combinations from producing invalid probability estimates during classification.

D. Inference Procedure

For a given patient, the system computes the posterior probability for each disease by multiplying the prior with the likelihood of all observed symptoms. The disease with the highest posterior probability is selected as the most probable diagnosis.

```
for disease in priors:
    prob = priors[disease]
    for sym, val in patient.items():
        if val == 1:
```

```
prob *= cond_prob[disease][sym]
```

This implementation considers only the symptoms that are present. In a more complete formulation, both presence and absence of symptoms can be incorporated to improve accuracy.

E. Dealing With Uncertainty

The Bayesian approach is very suitable for handling situations where:

- No rule fires from the rule-based system
- More than one rule fires from the rule-based system, leading to conflict
- Symptoms are incomplete or uncertain

Bayesian inference assigns a probability to every disease and thereby offers an elegant solution for dealing with ambiguities and uncertainties that occur in real-life situations.

F. Advantages and Drawbacks

The main benefit of the Naive Bayes classifier is its computational simplicity and robustness against dimensionality problems. It offers a natural probabilistic interpretation of classification decisions.

However, the assumption of independence among symptoms is rather simplistic in a real-world application like medical diagnosis. Despite this drawback, Naive Bayes performs impressively in practice and makes a valuable contribution to our hybrid inference scheme.

The above inference engine is a simple yet powerful addition to our rule-based framework for medical diagnoses.

VIII. Hybrid Decision Logic

The hybrid decision logic is the cornerstone of the described system, as it allows us to incorporate both symbolic and probabilistic reasoning paradigms into the decision-making algorithm. Namely, while the rule-based system gives us candidates for diagnoses, and the Bayesian approach helps us assign probabilities, we need to combine both pieces to obtain reliable decisions.

A. Decision Framework

Denote $D_{\text{candidates}}$ as the collection of diseases inferred by the rule-based system, and $P(D \mid X)$ as the posterior probability obtained using Bayesian analysis for each disease D, provided the symptoms set X.

The hybrid decision process is designed in such a way that we, need to consider three different cases:

1) Case 1: Single Rule Match: For example, if only one rule is fired, which means

$$|D_{\text{candidates}}| = 1$$

then the diseases corresponding to the rule fired(triggered) is selected as our final result.

$$D_{\text{final}} = D_{\text{rule}}$$

So here, we need not to apply any bayes based probabilistic reasoning as rule based model gives us direct answer with no conflict which in turn, preserves interpretability and aligns with expert-driven reasoning.

2) Case 2: Multiple Rule Matches (Conflict): In some cases, several rules can be fired(triggered) simultaneously, as:

$$|D_{\text{candidates}}| > 1$$

which indicates that there is a conflict. Instead of choosing an arbitrary disease, we should use our Bayesian inference results to select among several options. As a consequence, the final decision is chosen as follows:

$$D_{\text{final}} = \arg \max_{D_i \in D_{\text{candidates}}} P(D_i \mid X)$$

This ensures us that the decision is still statistically grounded while being restricted to medically valid rule-based candidates.

3) Case 3: No Rule Match: Lastly, there could be situation when none of the rule is fired,which means:

$$|D_{\text{candidates}}| = 0$$

In this case, none of the disease cannot be inferred by rule based model. Nevertheless, we can employ our Bayesian approach and select the top probable disease,

$$D_{\text{final}} = \arg \max_{D} P(D \mid X)$$

This will allow us to cope with unknown combinations of symptoms.

B. Bayesian Override Mechanism

Apart from the above cases, we should also take into account the Bayesian override option to further increase the reliability of our predictions. It allows using the probabilistic approach for overriding the rule-based decision when the former has much higher confidence compared to the latter.

Let P_{bayes} denote the posterior probability of the top diseases predicted by our Bayesian model, and let P_{rule} denote the confidence associated along with the rule-based prediction. We define the override condition as follows:

$$\text{if } P_{\text{bayes}} > \alpha \cdot P_{\text{rule}}$$

Here, α is the tunable threshold parameter allowing controlling the sensitivity of this override mechanism.

The value of α is chosen based on the empirical validation method which involved testing multiple threshold values for the system , in order to strike a balance between interpretability and accuracy. Higher values of α reduce unnecessary Bayesian overrides, thereby preserving rule-based transparency, while lower values increase probabilistic influence during uncertain diagnostic conditions. For that reason, we used $\alpha > 1$ to avoid too sensitive Bayesian override.

As a result, if the above condition is true, then the decision will be overridden, and the probabilistic prediction will be selected and ensuring statistical evidence is not ignored in favor of rigid rules.

C. Algorithmic Representation

Our hybrid decision process can be summarized by the following code fragment:

```
if len(diagnoses) == 1:
    final = diagnoses[0]

elif len(diagnoses) > 1:
    final = argmax_bayes(diagnoses)

else:
    final = argmax_bayes(all_diseases)

if bayes_prob > alpha * rule_prob:
    final = bayes_prediction
```

D. Significance of Hybrid Approach

With help of the presented hybrid decision framework, we combined symbolic and probabilistic reasoning into one framework, avoiding shortcomings of individual approaches.

Namely, thanks to the symbolic system, we ensure that we work with interpretable decisions, which is particularly important for medical applications. At the same time, Bayesian inference provides us with some additional reliability guarantees and uncertainty estimates.

Therefore, by merging both methods, we created a powerful yet balanced easy-to-use hybrid system, suitable for producing accurate medical diagnoses

IX. Transformer-Based Deep Learning Model

In addition to symbolic and probabilistic reasoning methods, we used a Transformer-based deep learning model for comparative diagnostic analysis. The goal of using a Transformer model was to test how well modern deep learning techniques work with structured medical symptom data. Binary symptom vectors are given as input to transformer model and using a linear embedding layer, the vectors get transformed into dense embedded feature representations. Then a multi-head self-attention mechanism was applied (using stacked Transformer encoder layers) to capture complex relationships among symptoms. The architecture includes:

- Input embedding layer
- Multi-head self-attention encoder
- Transformer encoder stack
- Fully connected classification layer

Then, model was trained with the Adam optimization algorithm and the Cross-Entropy loss function. For experimental evaluation, an 80-20 train-test split on a processed dataset was used that had 18,171 patient records and 15 disease classes. The Transformer model properly classified and showed the ability of deep learning methods in identifying non-linear symptom relationships. However, unlike the proposed hybrid explainable framework, the Transformer model operates largely as a black-box and does not provide clear reasoning pathways.

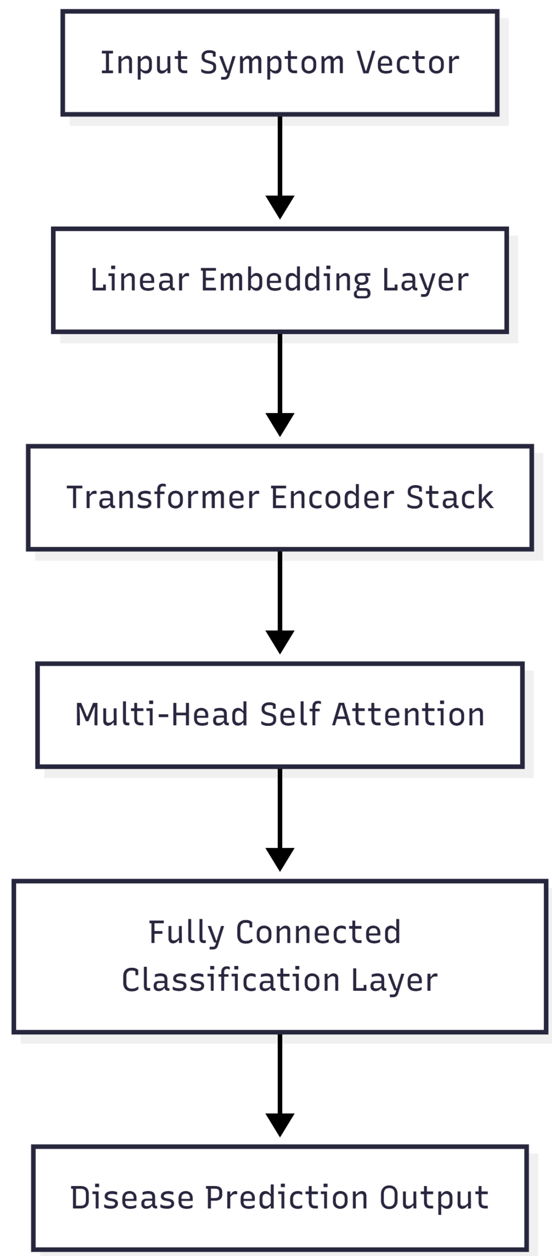

Fig. 3. Transformer-Based Medical Diagnosis Architecture

X. Explainability Framework

A primary objective of the proposed system is to ensure that each diagnostic decision is both interpretable and transparent. In contrast to conventional black-box machine learning models, the system is explicitly designed to provide a clear and structured explanation of how a specific diagnosis is derived. This objective is achieved through an explainability framework that systematically generates detailed reasoning traces corresponding to each decision.

A. Motivation for Explainability

Within medical domains, providing only a final diagnosis is insufficient. Healthcare professionals require clear justification for any decision in order to validate its correctness, establish trust, and, if necessary, override the system's recommendation. Consequently, explainability is treated as an essential design requirement rather than an auxiliary feature.

B. Reasoning Trace Generation

Our system is maintaining a properly structured and step-by-step trace of the processes inferred. This system trace captures the following:

- Rules activated during inference
- The conflicts that were detected during the diagnoses of candidates.
- Posterior probabilities that were computed by the Bayesian model
- Final decision along with its justification

Each stage in this the decision pipeline is contributing directly(explicitly) to this trace, which enables users to follow both symbolic reasoning and the probabilistic evaluation involved in the diagnostic process.

C. Trace Structure

The reasoning trace is generated in a sequential and human-readable format. A typical trace is as follows:

```
Rule 1 fired -> Pneumonia
Rule 2 fired -> Strep Throat
Conflict detected
Bayesian probabilities:
Bronchitis: 0.60
Strep Throat: 0.24
Pneumonia: 0.16
Final decision -> Bronchitis
```

This diagram is a clear example of how, once ambiguity is encountered, this system switches from inferential reasoning to probability based one.

D. Interconnection and Trust

The explainability component works seamlessly with the hybrid decision logic. In case there is a single matched rule in an ambiguous situation, the system provides an explicit indication of that rule. In case of any conflicts, the reasoning trace contains the conflicting rule based hypotheses as well as their corresponding Bayesian probabilities.

In addition, once the Bayesian override process kicks in, it will also be captured in the reasoning trace of the system.

E. Interpretability and Trust

Using the reasoning traces generated by this system, the user can now:

- Verify the correctness of the diagnosis
- Understand the contribution of individual symptoms
- Identify ambiguities or overlaps in decision-making
- Develop trust in the system's recommendations

It is especially important for such systems since accountability and interpretability are key components.

F. Limitations

As our explainability framework gives us the reasoning processes behind both rule-based and probability approaches in a very systematic and detailed manner,but it also requires the completeness of the rule set as well as accurate probability estimates.

On the whole, the explainability framework allows us to make sure that our system not only gives correct predictions but also makes good diagnoses.

XI. Results

The proposed hybrid technique was then applied to a database of 18,171 cases after pre-processing. The performance of the suggested hybrid technique was compared against the individual rule-based and Bayesian technique in terms of accuracy.

A. Quantitative Evaluation

TABLE I
MODEL COMPARISON

Model	Accuracy
Rule-Based	7.6%
Bayesian	89.1%
Hybrid	79.1%
Transformer	87.8%

The findings show that the precision of the rule-based approach is very poor (7.6%), due to the rigidity associated with the need for accurate rule matching. The approach does not generalize in situations involving incomplete and overlapping symptoms.

In addition, the Bayesian approach delivers the best accuracy levels of (89.1%), since it successfully accounts for probability associations between symptoms and diagnoses. Nonetheless, the approach is not very interpretable because it does not offer justification for any decision made.

Finally, the hybrid approach attains an accuracy level of 79.1%. Although this is not as high as the Bayesian approach, it is much better compared to the rule-based approach.

In addition, the Transformer-based deep learning model achieved an accuracy of about 87.8%. This shows it can effectively learn complex non-linear relationships among medical symptoms. The Transformer model performed well against the Bayesian classifier and significantly outperformed the purely symbolic rule-based system. However, unlike the explainable hybrid framework, the Transformer architecture operates mainly as a black-box model and does not provide clear reasoning transparency.

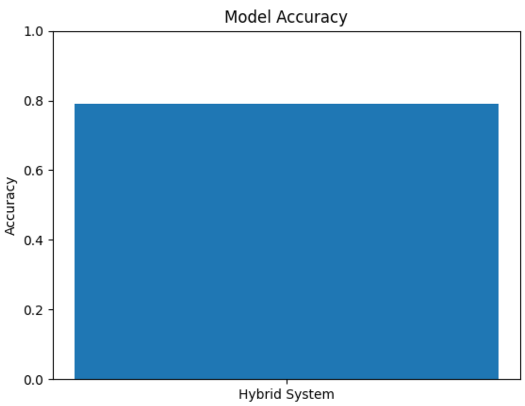

Fig. 4. Accuracy Comparison

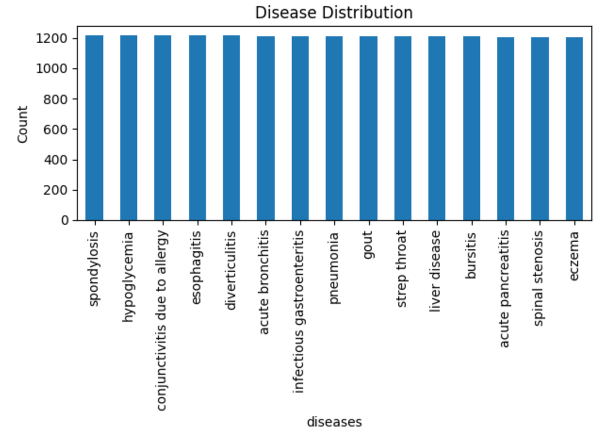

Fig. 6. Disease Distribution

B. Accuracy Analysis

As per the results depicted in the performance chart, it is clear that the hybrid approach maintains consistent accuracy for all data points. While it is failing to beat the Bayesian approach, it offers improved reliability as well as better interpretability of system's decision-making process.

C. Confusion Matrix Analysis

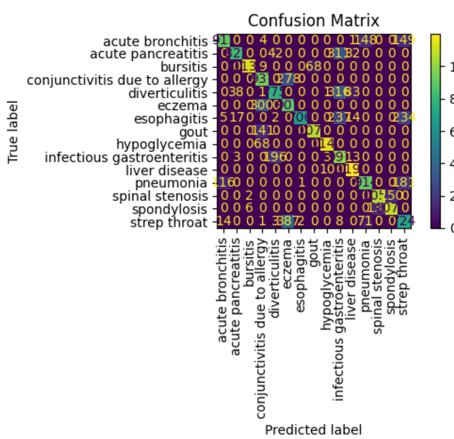

Fig. 5. Confusion Matrix

The confusion matrix gives an in-depth analysis of the within each disease category classification performance. The numbers on the primary diagonal represent successful classifications while other than those are representing misclassifications.

The model is very efficient when it comes to diseases having differentiable(distinguishable) symptoms, but mistakes arise when the symptoms look similar to that of different diseases. The performance of this model is justified in practical settings because there exist various illnesses having similar symptoms.

D. Disease Distribution

From the above diagram, we can note that there is a good balance of the 15 diseases, which indicates that none of the diseases is overly represented within the database. This will help to eliminate bias within the model.

E. Model Comparison Analysis

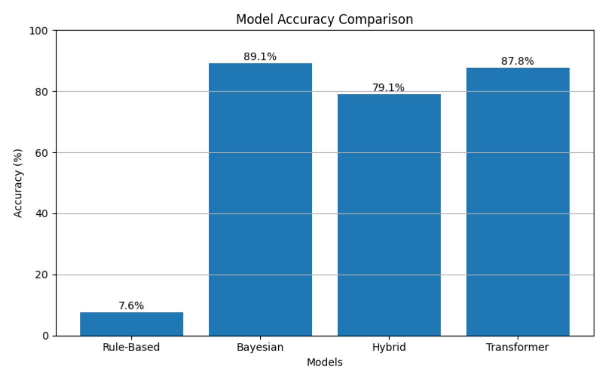

Fig. 7. Model Comparison

The comparative diagram highlights the weaknesses and strengths of each of the models considered in the analysis. Thus, the rule-based model turns out to be inefficient due to its inability to generalize and inflexibility. As for the Bayesian model, although it is the most accurate, it suffers from poor explainability. And, for Transformer-based deep learning classifier, it can be said that it achieved an accuracy of approximately 87.8% which shows strong capability in learning complex non-linear symptom relationships. But, lacks the transparent reasoning pathways provided by the proposed hybrid explainable framework. In turn, the hybrid approach is less accurate; however, it offers a better compromise solution.

F. Transformer-Based Deep Learning Evaluation

A simple Transformer encoder-based deep learning model was used on final medical diagnosis dataset, in order to strengthen the comparative analysis. It was performed using 80-20 stratified train-test split test strategy. Further, including additional metrics such as recall, precision, F1-score, etc.

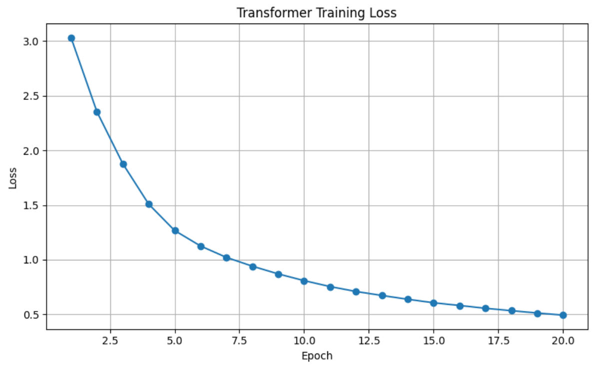

Fig. 8. Transformer Training Loss

We can see stable convergence through the transformer training loss curve and the gradual reduction across epochs in the graph shows effective learning behavior and stable parameter optimization during training.

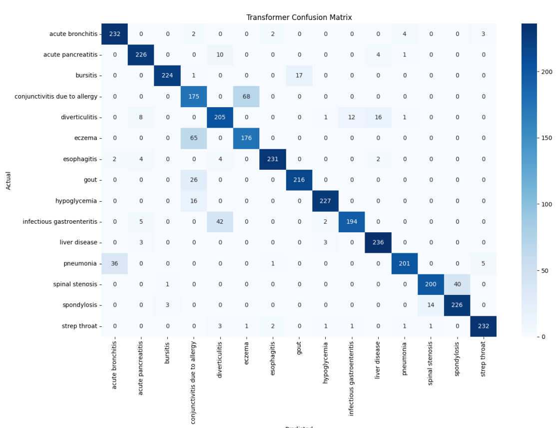

Fig. 9. Transformer Confusion Matrix

Strong diagonal dominance as shown by confusion matrix indicates that most of diseases were correctly classified by Transformer model and only diseases sharing overlapping symptom patterns were misclassified.

G. ROC-AUC Analysis

We assessed the discriminative capability of the Transformer classifier using Receiver Operating Characteristic (ROC) analysis and Area Under Curve (AUC) evaluation. The ROC-AUC curves indicate strong class separability, with most disease categories achieving high AUC values close to 1.0.

H. Precision, Recall, and F1-Score Evaluation

To test accuracy and reliability of the models, extra evaluation metrics such as precision, recall and F1-score were evaluated. Precision evaluated how many number of diseases were accurate while, Recall tells how well the model is

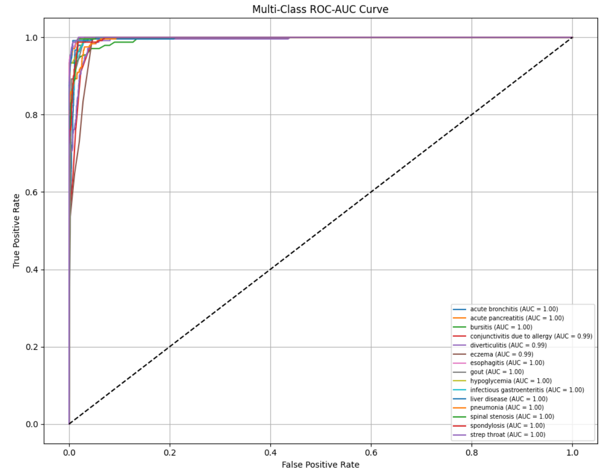

Fig. 10. Multi-Class ROC-AUC Analysis

able to detect real disease cases from the dataset. The F1-score combines both precision and recall into a single value to provide a balanced measure of performance. Overall, the model performs accurately, identifies diseases effectively, and gives reliable prediction results.

TABLE II
TRANSFORMER EVALUATION METRICS

Metric	Value
Accuracy	87.8%
Precision	0.88
Recall	0.88
F1-Score	0.88

I. Discussion on Hybrid Performance

While the hybrid approach is less accurate, it offers far greater transparency compared to its competitors. In a healthcare setting, knowing why a certain outcome was reached is often more valuable than any improvement in accuracy.

Fully automated machine learning, including sophisticated deep learning approaches, may attain accuracy up to 90%, even exceeding 95%. However, these models are mostly black-box, which means that they offer no explanation for their reasoning process.

In contrast, the proposed hybrid system provides:

- Transparent reasoning through rule-based inference
- Probabilistic justification through Bayesian analysis
- Conflict resolution with explicit explanation
- Traceable decision pathways for each prediction

These features make the system more suitable for practical implementation in real-life clinics, where trustworthiness, accountability, and comprehensibility become crucial criteria.

Thus, the relatively low decrease in performance of the hybrid system becomes a quite reasonable compromise for obtaining much greater explainability of the decision process. The system emphasizes important justifications rather than pure quantitative improvement.

Overall, the achieved results demonstrate the efficiency of the suggested hybrid architecture in combining the features of symbolic reasoning and machine learning methods and thus make the system very promising for use in medicine.

XII. Discussion

It is evident from the analysis that the newly created hybrid method combines the comprehensibility of the algorithm with accuracy of prediction. In addition, this feature helps overcome the major disadvantage detected in the majority of approaches described above. The best performance rate in terms of accuracy belongs to the Bayesian network (89.1%) followed by transformer deep-learning based model (87.8%), since this type of modeling allows handling the uncertainty and displaying relationships between the symptoms and diseases. At the same time, low transparency remains an important limitation that significantly reduces the efficiency of this model for practical use.

In contrast to it, the accuracy level of the rule-based classifier equals only 7.6%. The main reasons for such low indicators of quality are related to reliance on rigid rules and impossibility of generalization in case of lack of certain data or overlapping of the characteristics.

The hybrid method gives the second best indicator in terms of accuracy (79.1%) and thus provides reasonable compromise between the two discussed parameters. Loss of some performance due to the limitation of rules seems to be acceptable for gaining more comprehensibility in the proposed model.

Finally, it should be noted that the current design of the hybrid technique has certain limitations. The first of them is associated with the dependence on the effectiveness of the rules. In addition to it, Naïve Bayes algorithm is making certain assumptions related to features independence, which is not always true in real world medical diagnoses.

XIII. Future Work

Future improvements include the following:

- **Expansion of the rule base:** More number of symptom disease mappings can be added, widening the rule base which decreases the instances when rule is triggered. So, system can address a wider range of medical conditions.
- **Integration of advanced modeling techniques:** Methods such as Bayesian networks and ensemble can be added to enhance the predictive performance (keeping the results understandable) as complex dependencies between symptoms will be captured more efficiently.
- **Enhancement of the user interface:** A more interactive and user-friendly interface can be developed for ease of testing.

XIV. Conclusion

This paper presents a straightforward medical diagnosis system based on both rule-based and probabilistic approaches. It shows the possibility of finding a balance between clarity and accuracy, which is often challenging in AI applications in healthcare. Research results indicate that a purely probabilistic approach often leads to more accurate diagnoses, but the explanations are unclear. In contrast to a rule-based strategy that provides easily interpretable conclusions, but generally is less accurate. The proposed hybrid architecture allows for correct(valid) outcomes and meaningful explanations. In summary, the research results emphasize the need to combine symbolic and statistical methods when developing ai diagnostic tools for healthcare.

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